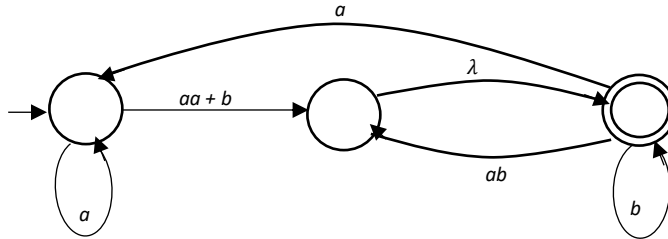


MET CS662 - Assignment #5

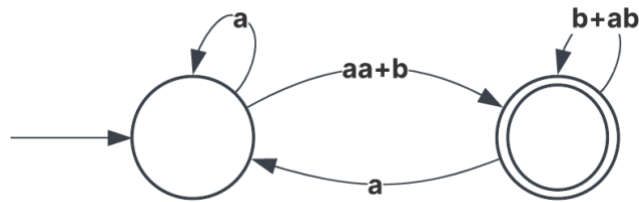
Joseph Rissman

U09545917

1. Consider the following generalized transition graph



- a. Find an equivalent generalized transition graph with only two states.



- b. What is the language accepted by this graph?

$$L(a^*(aa + b)((b + ab)^* + aa^*(aa + b))^*)$$

2. Construct right and left grammars for the language $L = \{a^n b^m \mid n \geq 2, m \geq 3\}$.

Right grammar:

$$\begin{aligned} S &\rightarrow aaA \\ A &\rightarrow aA \mid B \\ B &\rightarrow bB \mid bbb \end{aligned}$$

Left grammar:

$$\begin{aligned} S &\rightarrow Abbb \\ A &\rightarrow Ab \mid B \\ B &\rightarrow Ba \mid aa \end{aligned}$$

3. Construct a right linear grammar for the language $L((aab^*ab)^*)$.

Right linear grammar:

$$\begin{aligned} S &\rightarrow aaA \mid \lambda \\ A &\rightarrow bA \mid abS \end{aligned}$$

Left linear grammar:

$$\begin{aligned} S &\rightarrow Aab \mid \lambda \\ A &\rightarrow Ab \mid Saa \end{aligned}$$

4. Find a regular grammar that generates the language on $\Sigma = \{a, b\}$ consisting of all strings with no more than three a 's.

$$\begin{aligned} S &\rightarrow bS|aA|\lambda \\ A &\rightarrow bA|aB|\lambda \\ B &\rightarrow bB|aC|\lambda \\ C &\rightarrow bC|\lambda \end{aligned}$$

5. Find a regular grammar for the language $L = \{a^n b^m \mid n + m \text{ is even}\}$

$$\begin{aligned} S &\rightarrow aaS|abA|A \\ A &\rightarrow bbA|\lambda \end{aligned}$$

6. The symmetric difference of two sets S_1 and S_2 is defined as

$$S_1 \ominus S_2 = \{x \mid x \in S_1 \text{ or } x \in S_2, \text{ but } x \text{ is not in both } S_1 \text{ and } S_2\}$$

Show that the family of regular languages is closed under symmetric difference.

$$S_1 \ominus S_2 = (S_1 \cup S_2) \cap \overline{(S_1 \cap S_2)}$$

Supposing that S_1 and S_2 are regular languages, then $S_1 \ominus S_2$ must be a regular language because regular languages are closed under union, intersection, and complement.

7. Suppose we know that $L_1 \cup L_2$ is regular and that L_1 is finite. Can we conclude from this that L_2 is regular?

All finite languages are regular, so L_1 is a regular language. We can define $L_3 = L_1 - L_2$, the set of words in L_1 but not L_2 , which must be finite because it is a subset of a finite language. Therefore, L_3 must also be regular. Finally,

$$L_2 = (L_1 \cup L_2) - L_3$$

proves that L_2 is regular because regular languages are closed under set difference.

8. Which of the following are true for all regular languages and all homomorphisms?

a) $h(L_1 \cup L_2) = h(L_1) \cup h(L_2)$

This statement is true for all regular languages and all homomorphisms.

Let ω be a string in $h(L_1 \cup L_2)$.

There exists a string $x \in L_1 \cup L_2$ such that $h(x) = \omega$.

If $x \in (L_1 \cup L_2)$ then $x \in L_1$ or $x \in L_2$ or $x \in (L_1 \cap L_2)$.

If $x \in L_1$, then $h(x) \in h(L_1)$. If $x \in L_2$, then $h(x) \in h(L_2)$.

Therefore, $h(x) = \omega \in h(L_1) \cup h(L_2)$.

Conversely, if $\omega \in h(L_1) \cup h(L_2)$, then $\omega \in h(L_1)$ or $\omega \in h(L_2)$.

This implies there exists a string $x \in L_1$ or $x \in L_2$ such that $h(x) = \omega$.

Thus, $x \in (L_1 \cup L_2)$, $\omega \in h(L_1 \cup L_2)$. Therefore, $h(L_1 \cup L_2) = h(L_1) \cup h(L_2)$

b) $h(L_1 \cap L_2) = h(L_1) \cap h(L_2)$

This statement is not true for all regular languages and all homomorphisms.

Let $\Sigma = \{a, b\}$, $L_1 = \{ab\}$, $L_2 = \{ba\}$, and $h(a) = x$, $h(b) = x$.

$L_1 \cap L_2 = \emptyset$, so $h(L_1 \cap L_2) = \emptyset$.

$h(L_1) = \{xx\}$, $h(L_2) = \{xx\}$, so $h(L_1 \cap L_2) = \{xx\}$.

Therefore, $h(L_1 \cap L_2) \neq h(L_1) \cap h(L_2)$

c) $h(L_1 L_2) = h(L_1)h(L_2)$

This statement is true for all regular languages and all homomorphisms.

Let $\omega \in h(L_1 L_2)$. There is a string $x \in L_1 L_2$ such that $h(x) = \omega$.

Since $x \in L_1 L_2$, then $x = yz$, where $y \in L_1$ and $z \in L_2$.

Thus $h(x) = h(yz) = h(y)h(z)$.

Since $y \in L_1$, $h(y) \in h(L_1)$. Since $z \in L_2$, $h(z) \in h(L_2)$.

Thus $h(y)h(z) \in h(L_1)h(L_2)$.

Since $\omega = h(x) = h(yz) = h(y)h(z)$, $\omega \in h(L_1)h(L_2)$.

Therefore, $h(L_1 L_2) = h(L_1)h(L_2)$

Conversely, let $\omega \in h(L_1)h(L_2)$.

Then $\omega = uv$, where $u \in h(L_1)$ and $v \in h(L_2)$.

Therefore, there is a string $y \in L_1$, such that $h(y) = u$, and a string $z \in L_2$, such that $h(z) = v$.

Thus $\omega = h(y)h(z) = h(yz)$.

Since $y \in L_1$, and $z \in L_2$, $yz \in L_1 L_2$. Thus $\omega = h(yz) \in h(L_1 L_2)$.

Therefore, $h(L_1 L_2) = h(L_1)h(L_2)$.

9. If L is a regular language, prove that $L_1 = \{uv \mid u \in L, |v| = 2\}$ is also regular.

If L is a regular language, then there exists some DFA that recognizes L . Suppose the DFA is expressed as $M = (Q, \Sigma, \delta, q_0, F)$ where $L(M)$. A new automaton M_1 can be constructed to recognize L_1 as follows. Add transitions to each final state that will produce a new state q_1 for any input symbol:

$$\delta_1(q \in F, \Sigma) = q_1$$

Also add transitions from q_1 to a new final state q_f for any input symbol:

$$\delta_1(q_1, \Sigma) = q_f$$

The new transitions combined mean that there exists a transition from each of the original final states F to the new final state q_f with input words of length two:

$$\delta_1^*(q \in F, w) = q_f \text{ where } |w| = 2$$

Change all the original final states to non-final and make the new final state final:

$$F_1 = \{q_f\}$$

The resulting automaton $M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$ is an NFA where:

$$Q_1 = Q + \{q_1, q_f\}$$

$$\delta_1 = \delta + \delta^*(q \in F, v)$$

$$F_1 = \{q_f\}$$

Because $L_1(M_1)$ and M_1 is an NFA, then L_1 must be a regular language.

